

Homework 4

Simplicial Sets and Singular Chains

Math 615: Algebraic Topology I
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Use the constructions and results of Lecture 4. Give proofs and justify all asserted naturality statements. Unless a quotient is explicitly specified, retain all simplices, including degenerate ones. The definitions below fix the notation; the exercises begin on page 2.

Definitions and Notation

Functor Categories and Presheaves. The category $\text{Fun}(\mathcal{C}, \mathcal{D})$ has functors $\mathcal{C} \rightarrow \mathcal{D}$ as objects and natural transformations as morphisms. A natural transformation $\theta : F \rightarrow G$ consists of morphisms $\theta_c : F(c) \rightarrow G(c)$ satisfying $G(f)\theta_c = \theta_{c'}F(f)$ for every $f : c \rightarrow c'$. The opposite category $\mathcal{C}^{\text{op}}$ reverses all arrows of $\mathcal{C}$. Write

$$\mathcal{P}(\mathcal{C}; \mathcal{D}) = \text{Fun}(\mathcal{C}^{\text{op}}, \mathcal{D}), \quad \mathcal{P}(\mathcal{C}) = \mathcal{P}(\mathcal{C}; \text{Set}).$$

Objects of $\mathcal{P}(\mathcal{C})$ are called *presheaves of sets*. Here Set , Ab , and Top denote the categories of sets, abelian groups, and topological spaces, with functions, homomorphisms, and continuous maps respectively. The notation $\text{Mor}_{\mathcal{C}}(c, c')$ means the set of morphisms in $\mathcal{C}$.

The Simplex Category and Simplicial Objects. The category Δ has objects $[n] = \{0 < \dots < n\}$, $n \geq 0$, and weakly increasing maps as morphisms. A simplicial set is a presheaf $K \in \mathcal{P}(\Delta)$; write $K_n = K([n])$ and $\alpha^* = K(\alpha) : K_n \rightarrow K_m$ for $\alpha : [m] \rightarrow [n]$. An element of K_n is an *n-simplex*; vertices, edges, and triangles are simplices of degrees zero, one, and two. A simplicial map $f : K \rightarrow L$ is a natural transformation, so $f_m \alpha^* = \alpha^* f_n$. A simplicial abelian group is an object of $\mathcal{P}(\Delta; \text{Ab})$: its restriction maps are homomorphisms.

Faces, Degeneracies, and the Standard Simplex. For $n \geq 1$ and $0 \leq i \leq n$, let $\delta^i : [n-1] \hookrightarrow [n]$ be the injection omitting i . For $n \geq 0$ and $0 \leq i \leq n$, let $\sigma^i : [n+1] \twoheadrightarrow [n]$ be the surjection identifying i and $i+1$. Thus

$$\delta^i(k) = \begin{cases} k & k < i, \\ k+1 & k \geq i, \end{cases} \quad \sigma^i(k) = \begin{cases} k & k \leq i, \\ k-1 & k > i. \end{cases}$$

The induced maps $d_i = (\delta^i)^* : K_n \rightarrow K_{n-1}$ and $s_i = (\sigma^i)^* : K_n \rightarrow K_{n+1}$ are the *face* and *degeneracy* maps. For $n > 0$, a simplex of K_n is *degenerate* if it lies in some $s_i(K_{n-1})$, and *nondegenerate* otherwise; every vertex is nondegenerate. The standard simplicial n -simplex is

$$\Delta_m^n = \text{Mor}_{\Delta}([m], [n]), \quad \beta^*(\alpha) = \alpha \circ \beta.$$

A map $\alpha : [m] \rightarrow [n]$ also induces a map $\Delta^m \rightarrow \Delta^n$ by postcomposition.

Barycentric Coordinates. The standard topological n -simplex is

$$|\Delta^n| = \{(t_0, \dots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, t_0 + \dots + t_n = 1\},$$

with the subspace topology from $\mathbb{R}^{n+1}$. Its vertices are the standard basis vectors $e_0, \dots, e_n$. Each point has the unique expression $t_0 e_0 + \dots + t_n e_n$; the coefficients t_i are its *barycentric coordinates*. For example, $(1/2, 1/3, 1/6)$ is a point of $|\Delta^2|$, while $(1/2, 1/2, 0)$ lies on the edge opposite e_2 . The barycenter has every coordinate $1/(n+1)$.

For a weakly increasing map $\alpha : [m] \rightarrow [n]$, the associated affine map $\alpha_* : |\Delta^m| \rightarrow |\Delta^n|$ sends e_i to $e_{\alpha(i)}$:

$$(\alpha_* t)_j = \sum_{\alpha(i)=j} t_i \quad (\text{an empty sum is } 0).$$

In particular, $(\delta^i)_*$ inserts a zero coordinate and $(\sigma^i)_*$ adds adjacent coordinates $t_i + t_{i+1}$. A formula “in barycentric coordinates” is a formula in these t_i .

Realization and Singular Simplices. The geometric realization of K is the quotient space

$$|K| = \left(\coprod_{n \geq 0} K_n \times |\Delta^n| \right) / \sim, \quad (\alpha^* x, t) \sim (x, \alpha_* t),$$

where each K_n is discrete and the displayed pairs generate the equivalence relation. Write $[x, t]$ for the class of (x, t) . The characteristic map of $x \in K_n$ is $\chi_x : |\Delta^n| \rightarrow |K|$, $t \mapsto [x, t]$. A *singular n -simplex* in X is a continuous map $\tau : |\Delta^n| \rightarrow X$. These form the simplicial set $\text{Sing } X$, with $\alpha^* \tau = \tau \circ \alpha_*$. The realization–singular adjunction is the natural bijection

$$\text{Mor}_{\text{Top}}(|K|, X) \cong \text{Mor}_{\mathcal{P}(\Delta)}(K, \text{Sing } X).$$

Its unit $\eta_K : K \rightarrow \text{Sing } |K|$ sends x to χ_x .

Subsets, Boundaries, and Quotients. A simplicial subset $L \subseteq K$ consists of subsets $L_n \subseteq K_n$ preserved by every restriction map. The boundary $\partial \Delta^n \subseteq \Delta^n$ consists in degree m of the nonsurjective maps $[m] \rightarrow [n]$. When each L_n is nonempty, the quotient K/L identifies all elements of L_n to one element in each degree; its operators are induced from those of K . The notation $B \amalg_A C$ denotes a *pushout*: an object receiving maps from B and C agreeing on A , universal among such objects. Coproducts $K \amalg L$, products $K \times L$, and pushouts of simplicial sets are computed degree by degree in sets.

Exercise 1. Standard Simplices and Simplicial Operators

Represent an m -simplex of Δ^n by a weakly increasing sequence $(a_0, \dots, a_m)$ with $0 \leq a_i \leq n$.

1. Describe every face and degeneracy map on these sequences. Verify $d_i d_j = d_{j-1} d_i$ for $i < j$, and all three cases of the identity for $d_i s_j$, by tracking the entries deleted or repeated.
2. Prove that a simplex is nondegenerate precisely when its entries are strictly increasing. Count the m -simplices and the nondegenerate m -simplices of Δ^n . List the edges of Δ^2 , distinguishing its degenerate and nondegenerate edges.
3. For $x \in K_n$, define a map $\bar{x} : \Delta^n \rightarrow K$ by $\bar{x}_m(\alpha) = \alpha^* x$. Prove that evaluation at $\text{id}_{[n]}$ gives a natural bijection $\text{Mor}_{\mathcal{P}(\Delta)}(\Delta^n, K) \cong K_n$. Under this bijection, identify precomposition by the maps of standard simplices induced by δ^i and σ^i .

Exercise 2. Singular Simplices and Degeneracy

The homeomorphism $[0, 1] \rightarrow |\Delta^1|$ is $u \mapsto (1 - u, u)$. For a space X , recall that $(\text{Sing } X)_n = \text{Mor}_{\text{Top}}(|\Delta^n|, X)$.

1. For a singular triangle $\tau : |\Delta^2| \rightarrow X$, write formulas for $d_0 \tau$, $d_1 \tau$, $d_2 \tau$ as paths parametrized by $u \in [0, 1]$. Identify the initial and terminal vertex of each path.
2. For a path $\gamma : [0, 1] \rightarrow X$, write $s_0 \gamma$ and $s_1 \gamma$ in barycentric coordinates, and compute all three faces of each. Specify which resulting paths are constant.
3. Prove that a singular edge is degenerate if and only if it is constant. Give an explicit nonconstant path in $\mathbb{R}$ whose two endpoints are equal. Explain why it is nevertheless nondegenerate.
4. Describe the image in degrees zero and one of the unit $\eta_{\Delta^1} : \Delta^1 \rightarrow \text{Sing } |\Delta^1|$. Give a singular vertex and a singular edge outside the respective images. Deduce that this unit is not an isomorphism of simplicial sets.

Exercise 3. Realization, Quotients, and Loops

Let $S = \Delta^1 / \partial \Delta^1$ be the degreewise quotient collapsing the simplicial subset generated by the two vertices to a single vertex. Write v for its vertex and e for its nondegenerate edge.

1. Describe S_m in terms of weakly increasing sequences of zeros and ones. Prove that v and e are its only nondegenerate simplices. List all simplices in degrees one and two, including degeneracies.
2. Use the realization–singular adjunction to prove that realization preserves pushouts. Apply this to $S = \Delta^1 \amalg_{\partial \Delta^1} \Delta^0$ to identify $|S|$ with $[0, 1]/(0 \sim 1)$, including its topology. Explain why the simplicial quotient introduces no extra geometric cells.
3. Prove that a simplicial map $S \rightarrow K$ is equivalent to the choice of $x \in K_0$ and $a \in K_1$ with $d_0 a = d_1 a = x$. Apply this to $K = \text{Sing } X$ and identify such maps with pairs consisting of a point $x \in X$ and a continuous loop $\gamma : [0, 1] \rightarrow X$ based at x . These are actual loops, not homotopy classes.
4. For this correspondence, write explicitly the associated continuous map $|S| \rightarrow X$. Check that it agrees with the adjunction formula the formula $F([z, t]) = g_n(z)(t)$ for a simplicial map $g : S \rightarrow \text{Sing } X$.

Exercise 4. Free Simplicial Abelian Groups

For a set T , the *free abelian group* $\mathbb{Z}[T]$ consists of finite formal sums $\sum_{t \in T} n_t t$ with $n_t \in \mathbb{Z}$. A function $T \rightarrow U(B)$ to the underlying set of an abelian group extends uniquely to a homomorphism $\mathbb{Z}[T] \rightarrow B$; this is the free–forgetful adjunction. For $K \in \mathcal{P}(\Delta)$, put $\mathbb{Z}[K]_m = \mathbb{Z}[K_m]$, with restriction maps obtained by extending those of K linearly. A generator associated to $x \in K_m$ is denoted by the same symbol x . Let $A \in \mathcal{P}(\Delta; \text{Ab})$, and let $U(A)$ be its underlying simplicial set, obtained by forgetting the group structure in each degree. For an adjunction $F \dashv U$, its unit $\eta : \text{id} \rightarrow UF$ and counit $\epsilon : FU \rightarrow \text{id}$ correspond to identity morphisms under the adjunction bijections. Their triangle identities are $\epsilon_{F(K)} \circ F(\eta_K) = \text{id}_{F(K)}$ and $U(\epsilon_A) \circ \eta_{U(A)} = \text{id}_{U(A)}$.

1. Combine the free–forgetful adjunction with Exercise 1 to construct a natural group isomorphism

$$\text{Mor}_{\mathcal{P}(\Delta; \text{Ab})}(\mathbb{Z}[\Delta^n], A) \cong A_n.$$

Describe the homomorphism associated to $a \in A_n$ on every generator α in every degree. The group structure on the left is pointwise addition of simplicial homomorphisms.

2. For S from Exercise 3, identify $\text{Mor}_{\mathcal{P}(\Delta; \text{Ab})}(\mathbb{Z}[S], A)$ with the subgroup

$$\{(x, a) \in A_0 \times A_1 : d_0 a = d_1 a = x\}.$$

Describe directly the values of the associated homomorphism on $s_0 v$, $s_0 e$, and $s_1 e$.

3. Write the unit $K \rightarrow U\mathbb{Z}[K]$ and counit $\mathbb{Z}[U(A)] \rightarrow A$. Verify the two triangle identities on generators. In $\mathbb{Z}[U(A)]_n$, exhibit a nonzero element sent to zero by the counit; explain why 0 is not the zero element of this free abelian group.
4. Prove naturally that $\mathbb{Z}[K \amalg L] \cong \mathbb{Z}[K] \oplus \mathbb{Z}[L]$. For $K = L = \Delta^0$, show that the canonical map induced by the two projections, $\mathbb{Z}[K \times L] \rightarrow \mathbb{Z}[K] \times \mathbb{Z}[L]$, is not an isomorphism.

Exercise 5. Alternating Faces and Degenerate Chains

For a simplicial abelian group A , let $C_n(A) = A_n$ and $\partial_n = \sum_{i=0}^n (-1)^i d_i$ for $n > 0$. For a simplicial set K , abbreviate $C_\bullet(\mathbb{Z}[K])$ to $C_\bullet(K; \mathbb{Z})$. Set $C_n = 0$ for $n < 0$ and $\partial_0 = 0$. A chain complex has $\partial_{n-1} \partial_n = 0$; a chain map commutes with the differentials. A subcomplex $D_\bullet \subseteq C_\bullet$ means $\partial_n(D_n) \subseteq D_{n-1}$, so the quotient groups carry an induced differential. The *unnormalized* complex retains all simplices; the quotient by degenerate chains is called the *normalized quotient complex*.

1. Expand $\partial_2 \partial_3$ and pair its twelve terms using the face identities. Explain how the same pairing proves $\partial^2 = 0$ in every degree. Show that a simplicial homomorphism induces a chain map.
2. Put $D_0(A) = 0$ and $D_n(A) = \sum_{j=0}^{n-1} \text{im}(s_j : A_{n-1} \rightarrow A_n)$ for $n > 0$. For $a \in A_{n-1}$, expand $\partial_n(s_j a)$ using the mixed face–degeneracy identities. Show that the two terms not visibly in a degeneracy image cancel. Deduce that $D_\bullet(A)$ is a subcomplex, treating $n = 1$ explicitly.
3. For $A = \mathbb{Z}[K]$, prove that $D_n(A)$ is freely generated by the degenerate n -simplices. Hence identify $C_n(A)/D_n(A)$ with the free abelian group on the nondegenerate n -simplices, and describe its induced differential. You need not prove that this quotient preserves homology.
4. Compute the *unnormalized* complex $C_\bullet(\Delta^0; \mathbb{Z})$ and its homology in all degrees. For its vertex v , explain explicitly why $s_0 v$ is a nonzero cycle but represents zero in homology. Compare with the quotient complex from the preceding part.

Exercise 6. Components and a First Circle Calculation

Write $H_n(K; \mathbb{Z}) = \ker(\partial_n) / \text{im}(\partial_{n+1})$ for the homology of $C_\bullet(K; \mathbb{Z})$: elements of the kernel are *cycles* and elements of the image are *boundaries*. Define $H_n^{\text{sing}}(X; \mathbb{Z}) = H_n(\text{Sing } X; \mathbb{Z})$. A path from x to y is a continuous map $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = x$, $\gamma(1) = y$; it is a loop based at x if $y = x$. Path components are equivalence classes under the existence of a path; connected components are maximal connected subsets. Let $\pi_0(K)$ denote the quotient of K_0 by the equivalence relation generated by $d_1 a \sim d_0 a$ for every $a \in K_1$.

1. Prove that the map taking each vertex to its equivalence class induces a natural isomorphism

$$H_0(K; \mathbb{Z}) \cong \mathbb{Z}[\pi_0(K)].$$

Show that the kernel of $\mathbb{Z}[K_0] \rightarrow \mathbb{Z}[\pi_0(K)]$ is generated by $d_0 a - d_1 a$.

2. Deduce naturally that $H_0^{\text{sing}}(X; \mathbb{Z}) \cong \mathbb{Z}[\pi_0^{\text{path}}(X)]$, where $\pi_0^{\text{path}}(X)$ is the set of path components. Explain why connected components cannot simply replace path components in this argument. No example of a connected, non-path-connected space is required.
3. If X is discrete, identify $\text{Sing } X$ and compute $H_n^{\text{sing}}(X; \mathbb{Z})$ for every $n \geq 0$, including $X = \emptyset$. Use the actual unnormalized differential.
4. For $S = \Delta^1 / \partial \Delta^1$, use the lists from Exercise 3 to compute $C_0(S; \mathbb{Z})$, $C_1(S; \mathbb{Z})$, $C_2(S; \mathbb{Z})$ and both differentials. Deduce $H_0(S; \mathbb{Z})$ and $H_1(S; \mathbb{Z})$, specifying a generator of H_1 . Work directly in the unnormalized complex; do not assume a comparison theorem with the singular homology of $|S|$.